

Calculate intervals across zero

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: What is the difference between -4 and 7? Copy or complete the first step.

2. The temperature is -6 C and rises by 9 C. What is it now?

3. How far is -12 from 5 on a number line?

4. Miro says: Miro counts both zero and the starting number when finding the interval from -4 to 7. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: What is the difference between -4 and 7? Copy or complete the first step.
Answer 11.
2. The temperature is -6 C and rises by 9 C. What is it now?
Answer 3 C.
3. How far is -12 from 5 on a number line?
Answer 17.
4. Miro says: Miro counts both zero and the starting number when finding the interval from -4 to 7. Is this correct? Explain.
Answer Count the jumps between numbers: 4 jumps to zero and 7 more to reach 7, giving 11.

Calculate intervals across zero

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. The temperature is -6°C and rises by 9°C . What is it now?

2. How far is -12 from 5 on a number line?

3. What number is halfway between -8 and 6 ?

4. Identify and correct this misconception: Miro counts both zero and the starting number when finding the interval from -4 to 7 .

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. The temperature is -6°C and rises by 9°C . What is it now?

Answer 3°C .

2. How far is -12 from 5 on a number line?

Answer 17 .

3. What number is halfway between -8 and 6 ?

Answer -1 .

4. Identify and correct this misconception: Miro counts both zero and the starting number when finding the interval from -4 to 7 .

Answer Count the jumps between numbers: 4 jumps to zero and 7 more to reach 7, giving 11.

5. Write a complete sentence explaining how you checked one answer.

Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Calculate intervals across zero

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. What number is halfway between -8 and 6?

2. Explain why this reasoning fails: Miro counts both zero and the starting number when finding the interval from -4 to 7.

3. Create a new example that tests the same idea as: How far is -12 from 5 on a number line?

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. What number is halfway between -8 and 6?

Answer -1.

2. Explain why this reasoning fails: Miro counts both zero and the starting number when finding the interval from -4 to 7.

Answer Count the jumps between numbers: 4 jumps to zero and 7 more to reach 7, giving 11.

3. Create a new example that tests the same idea as: How far is -12 from 5 on a number line?

Answer Answers vary. The example and solution must preserve the mathematical structure.

4. Find a second method or representation. Compare its efficiency with your first method.

Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

5. Write one always, sometimes or never statement about today's learning and justify it.

Answer Answers vary. A valid example and counterexample should support the classification.